



Design and service performance of FRP anchorage system: A review

Yanzhao Niu^{a,b,c}, Guijun Xian^{a,b,c}, Rui Guo^{a,b,c,*}

^a Key Lab of Structures Dynamic Behavior and Control (Ministry of Education), Harbin Institute of Technology, Harbin, 150090, China

^b Key Lab of Smart Prevention and Mitigation of Civil Engineering Disasters of the Ministry of Industry and Information Technology, Harbin Institute of Technology, Harbin, 150090, China

^c School of Civil Engineering, Harbin Institute of Technology, Heilongjiang, Harbin, 150090, China

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ABSTRACT

Fiber Reinforced Polymer (FRP) materials, with their excellent properties such as lightweight, high strength, and corrosion resistance, have become highly promising materials in modern engineering. However, to fully harness the advantages of FRP, a reliable anchorage system is crucial. This paper provides a comprehensive review of the research on FRP anchorage systems, focusing on static tensile, fatigue and durability properties, exploring the working mechanisms, failure modes and performance influencing factors of anchorage systems. The static tensile properties of anchorages are closely related to the type of anchorages, with mechanical, bonded, composite and self-anchored types anchorage systems each having their own strengths and weaknesses. Regarding fatigue performance, mechanical-type anchorages are prone to interfacial wear and slippage, while bonded and composite types anchorages exhibit shear and slippage failures, and self-anchored systems tend to suffer from localized fracture due to stress concentration. Design optimization and material improvements can help mitigate fatigue damage. Moreover, both FRP and the bonding medium are susceptible to performance degradation or electrochemical corrosion under complex environmental conditions. The bonding performance at the interfaces of anchorage components deteriorates under the combined effects of load and environmental factors, posing a significant threat to the long-term performance of FRP anchorage systems. This study provides valuable insights into the performance of FRP anchorage systems in engineering applications, driving their technical advancement and practical implementation.

1. Introduction

Fiber reinforced polymer (FRP) materials with the excellent mechanical properties, corrosion resistance and lightweight nature are gradually replacing traditional reinforcement materials in modern engineering, especially in complex environments where they offer significant application value [1–3]. However, to fully harness the advantages of FRP, a reliable anchorage system is crucial. Static tensile performance, fatigue resistance and long-term durability are key criteria for assessing the overall performance of anchorages [4]. As a crucial indicator of FRP anchorages, static tensile performance determines its structural load-bearing capacity and stability, particularly under multi-field coupling effects, which are vital for enhancing safety and long-term reliability in engineering applications [5]. While FRP offers distinct advantages in terms of corrosion resistance and aging resistance, the anchorages still face complex challenges related to fatigue damage and durability under long-term loading and environmental influences [6].

Fatigue damage mechanisms, interface degradation and environmental-mechanical coupling effects are key factors influencing the long-term reliability of FRP anchorage systems.

Current research still confronts two critical challenges. First, the design of anchorage systems of FRP necessitates a trade-off between short-term strength and long-term reliability. For instance, increasing anchorage length mitigates stress concentration but risks introducing additional potential interfacial defects. Second, the discrepancy between laboratory-controlled environments and real-world engineering service conditions leads to significant deviations in lifespan prediction models. To address these issues, future advancements should integrate interfacial chemistry, composite mechanics, and data-driven analytics, thereby establishing a multi-scale theoretical framework spanning from molecular bonding stability to structural life-cycle responses. Such interdisciplinary efforts are imperative to enable reliable applications of FRP anchorage systems in extreme environmental engineering scenarios. This paper systematically evaluates contemporary FRP anchorage systems,

* Corresponding author. 73 Huanghe Road, Nangang District, Harbin Institute of Technology, Harbin, 150090, China.

E-mail address: guorui@hit.edu.cn (R. Guo).

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presenting a critical review of their mechanical behavior under static tensile loading, cyclic fatigue conditions and prolonged environmental exposure. The synthesis of current research findings provides essential insights for optimizing anchorage system design to enhance structural safety and service life in practical engineering implementations.

2. Types of FRP anchorage systems

2.1. Mechanical-type anchorages

The mechanical-type anchorages have been successfully applied to steel cable structures for over half a century, providing valuable insights and experience for the design of FRP anchorage systems. A commonly used mechanical-type anchorage of FRP typically consists of an anchor barrel and wedges, as shown in Fig. 1a. The anchoring mechanism relies on the mechanical interlocking force generated by the deformation of the anchor barrel's internal cavity, which anchors FRP through the wedges [7,8]. This process requires the wedges to have sufficient compressive strength, while the anchor barrel must be capable of withstanding expansion-induced failure. To enhance the mechanical interlocking force and prevent sliding failure of the FRP, preloading is usually applied to the wedges before loading [9].

2.1.1. Split-wedge anchorage systems

To ensure even compressive stress on FRP and prevent stress concentration at the loading end of the anchorage, Sayed-Ahmed et al. [10, 11] proposed a split-wedge anchorage system (SWAS), as shown in Fig. 1b. A copper sleeve is placed between the steel wedges and FRP bar, ensuring that the steel wedges apply a more even compressive stress to FRP bar. Additionally, the internal cavity of the steel barrel and the external wall of the steel wedges are set at an angular difference to prevent stress concentration at the loading end of the anchorage. The modified system, with an 8 mm Leadline tendon, achieved a failure load of 114 kN (2268 MPa), surpassing 10 % of the tendon's guaranteed strength. Further, Al-Mayah et al. [12,13] replaced the copper sleeve with an aluminum one, significantly reducing FRP bar slippage. Fang et al. [14] added a plastic film between the steel barrel and steel wedges, increasing the wedge's slip and enhancing the mechanical bonding force

of the anchorage.

Al-Mayah et al. [15,16] introduced the curved wedge anchorage system (CWAS), which uses a contoured profile of the barrel and wedges to achieve uniform contact pressure and improved anchoring performance, as shown in Fig. 1c. The best results were achieved with a CWAS radius of 1900 mm, demonstrating a failure load of 152 kN and minimal displacement, with no significant effect from different tendon sizes, sleeve materials, or loading rates.

2.1.2. Non-metallic anchorage systems

Mechanical-type anchorage systems with the steel barrel and steel wedges are prone to electrochemical corrosion in alkaline environments, which has led to increased attention on the use of non-metallic anchorage components [18]. Taha et al. [19–21] developed a non-metallic wedge anchorage system (N-MWAS) made of ultra-high-performance concrete (UHPC), as shown in Fig. 2a. Due to the impracticality of using a thick non-metallic barrel, CFRP sheets were applied to reinforce and reduce its thickness. The system required a presetting load of 65 kN, with an average failure load of 100 kN (1989 MPa) in static tensile tests. Compared to metal anchorage systems, N-MWAS offers larger dimensions and lower cost. To reduce weight, Shaheen et al. [22,23] developed a new carbon fiber-reinforced reactive powder concrete (CFRRPC) for manufacturing the anchorage, which is an ultrahigh-strength concrete mixture with super-plasticized silica fume and 3 mm long carbon fibers. The anchor barrel was wrapped with CFRP sheets using a butt joint. The results showed that the anchorage had an average efficiency coefficient of 96 %. This anchor weighs about one-third as much as steel anchors (475 g), though its ultimate anchoring capacity is lower.

Terrasi et al. [24] proposed a sleeve-wedge anchorage system (SWAS). The three wedges were made of a thermoplastic polymer, polyphenylene sulfide (PPS), as shown in Fig. 2b. The static rupture load under laboratory conditions increased by 25 %, and the pretensioning level on-site could be increased by 50 %. Zhou et al. [25] proposed the small-sized variable-stiffness wedge (VSW) anchorage, which features a non-metallic wedge with a variable stiffness profile, where the elastic modulus decreases from the free end of the anchor to the loading end, as shown in Fig. 2c. Tensile test results show that the anchoring efficiency of the VSWs is 87 %.

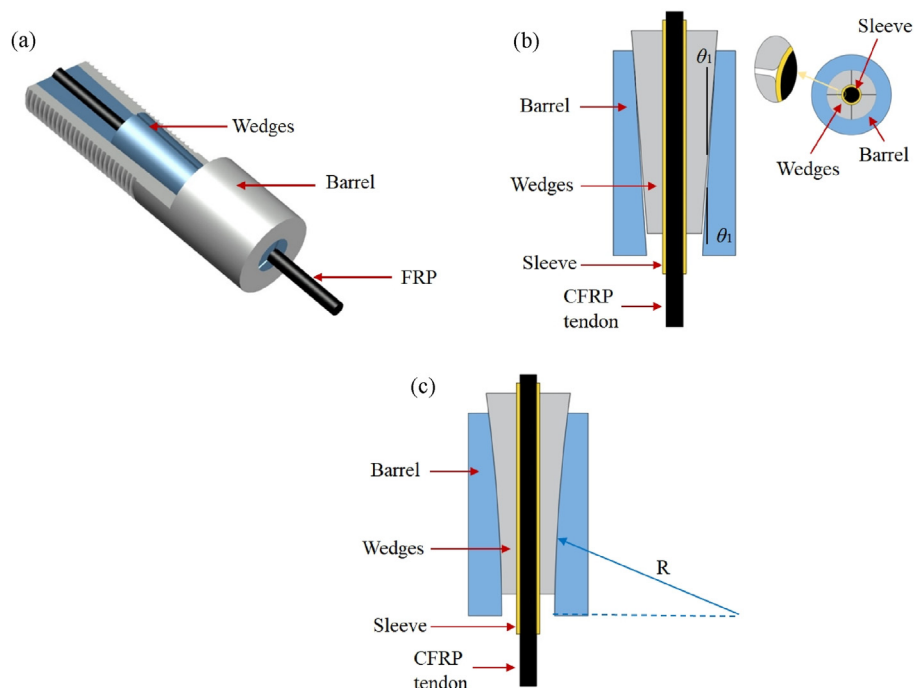


Fig. 1. Mechanical-type anchorages of (a) Commonly used anchorage, (b) SWAS and (c) CWAS [15,17].

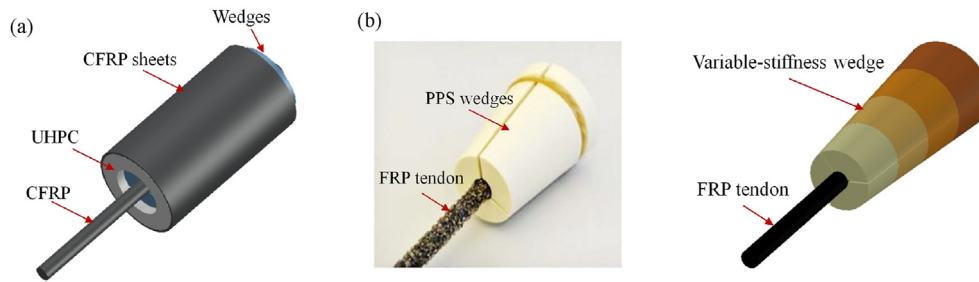


Fig. 2. Non-metallic anchorages of (a) N-MWAS, (b) SWAS (PPS wedges) and (c) VSW [19,24,25].

2.1.3. Prestressed-type anchorage systems

Prestressed-type anchorages typically apply preload to the anchorage through bolts, which allows for precise control over the anchoring force, and helps to enhance the stability and efficiency of the anchorage. The ability to control the anchoring force is crucial in maintaining the long-term reliability and safety of the system, especially in applications where the load conditions may vary over time.

Mohee et al. [26] used two steel plates to clamp steel wedges, anchoring CFRP plate. The clamping force was controlled by bolts on the steel plates, as shown in Fig. 3a. This novel anchorage can grip and prestress CFRP plate to its full tensile capacity of 168 kN without any premature failure. Ding et al. [27] proposed a novel lap joint design for CFRP plates, termed the preclamp lap joint, as shown in Fig. 3b. Three types of pre-clamp lap joints were tested, based on the characteristics of the clamping device: planar, curved, and thickened. By applying an appropriate bolt preload and adhesive method, both the curved pre-clamp single lap joint and the thickened pre-clamp joint could achieve a failure mode characterized by CFRP plate rupture, rather than failure at the joint itself. Duo et al. [28,29] proposed a novel variable curvature clamping anchor (VCCA) for multilayer CFRP plate cables, as shown in Fig. 3c. The results showed that the peak compressive stress was effectively transferred to the back of the anchorage, while the interfacial shear stress on CFRP plate remained low, less than one-sixth of the shear strength. The number of waveforms and clip thickness had little impact on peak compressive stress, while bolt preload, anchor length, ear plate thickness, and clamping plate thickness significantly affected axial stress distribution and peak compressive stress.

In conclusion, mechanical-type anchorages offer the advantages of easy installation without the need for adhesive curing, making them suitable for harsh environments and rapid construction requirements. However, the mechanical gripping mechanism can lead to stress concentration in FRP, posing a risk of localized damage. These anchors also demand high precision in manufacturing and installation processes, and may present durability concerns under long-term dynamic loading. Furthermore, mechanical-type anchorages are typically designed to secure single FRP tendon, making them unsuitable for anchoring FRP cables composed of multiple tendons.

2.2. Bonded-type anchorages

Bonded-type anchorages generally comprise three main components: FRP, bonding medium and anchor barrel. The bonding medium establishes a chemical bond with FRP, ensuring a secure and durable fixation [30]. Unlike anchorages that rely on physical friction or mechanical locking mechanisms, this type of anchorage depends on chemical adhesion [31,32].

2.2.1. Design of anchor barrel

As illustrated in Fig. 4, anchor barrels typically appear in three common shapes: straight, inner cone, and a combination of straight and inner cone, each designed to offer distinct advantages for various applications. The straight-barrel design is straightforward and easy to manufacture, making it a practical choice for many scenarios. In contrast, the inner cone-barrel significantly enhances anchorage efficiency [9]. The

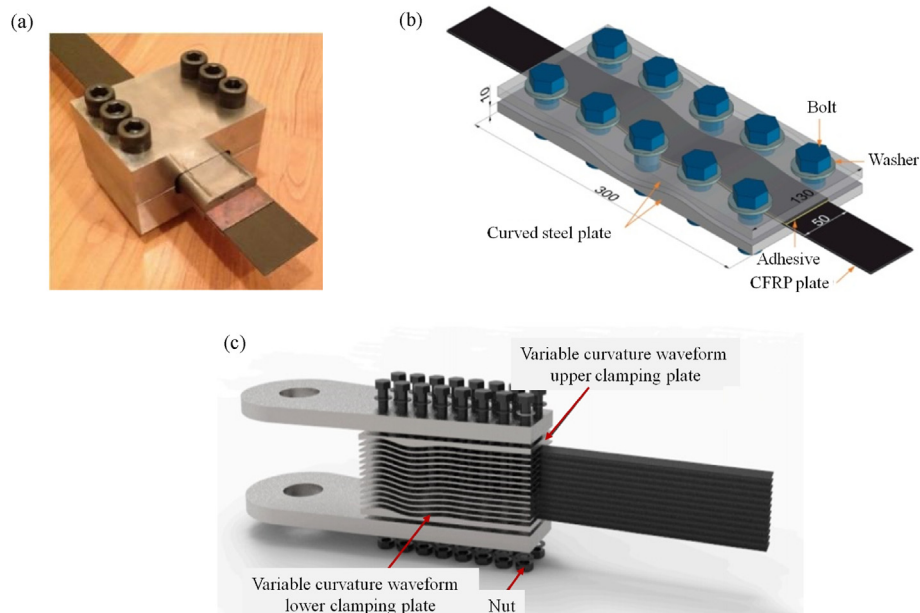


Fig. 3. Prestressed-type anchorages of (a) Steel bolts anchorage, (b) Lap joint anchorage and (c) VCCA [26–28].

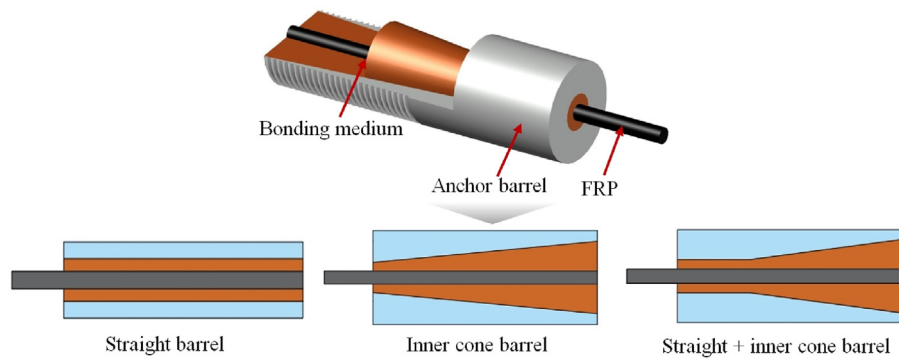


Fig. 4. Bonded-type anchorages.

hybrid design, which combines both straight and inner cone features, provides enhanced stability and load-bearing capacity, making it particularly suitable for demanding conditions.

Based on straight-barrel anchorage, Zhang et al. [33] found that bond stress at the tendon-adhesive interface is uneven, with the peak bond strength near the loading end, shifting towards the free end as the load increases. The stress distribution depends on anchorage length, with shorter lengths providing a more uniform distribution, while longer lengths lead to uneven stress and complicate transportation and installation. Currently, most bonded-type anchorages use an inner cone anchor barrel, which provides circumferential confinement to the bonding medium, enhancing bond strength and reducing the anchorage length compared to straight-barrel anchorages [34,35]. Mei et al. [36,37] proposed a straight plus inner cone barrel anchorage to address stress concentration at the inner cone anchorage loading end. Tensile tests showed that the anchorage exhibited high efficiency, with filamentary pull-off damage as the primary tendon failure mode.

2.2.2. Types of bonding medium

The properties of the bonding medium play a pivotal role in determining the anchoring performance. Variations in bonding materials can significantly influence bond strength, stress distribution, and the overall stability of the anchorage system, underscoring the importance of selecting an appropriate bonding medium to optimize anchoring efficiency. Wolff et al. [38] demonstrated successful anchoring of FRP bars using epoxy sand as the bonding medium, achieving remarkable anchoring performance with an efficiency exceeding 90 %. Similarly, Mochida et al. [39] adopted cement slurry as bonding medium, and the test results showed that the effect of anchoring length on bearing capacity of anchorage was significant. Zhang et al. [33] utilized cement grout as the bonding medium and developed a model to calculate bond stress and tensile capacity, identifying tendon and steel sleeve geometry, along with grout properties, as critical factors affecting stresses at the tendon-grout interface. Additionally, Nanni et al. [7,40] anchored FRP rods using epoxy resin as the bonding medium and compared their performance with wedges and spike anchors. Their findings revealed that epoxy resin anchorage delivered superior anchoring performance, with the potential for even higher efficiency through further optimization.

2.2.3. Variable stiffness bonded-type anchorage

To mitigate stress concentration at the loading end of the anchorage, researchers have adopted a variable-stiffness bonding medium. Specifically, the loading end of the anchorage employs low-modulus bonding media to fully accommodate transverse FRP deformation under tension, thereby avoiding radial compressive stress concentration. Conversely, the rear end of the anchorage utilizes high-modulus bonding media to convert residual expansion into interfacial friction through rigid constraints, enabling progressive load transfer. This innovative design not only reduces stress peaks but also significantly enhances anchoring efficiency [41].

Meier et al. [42,43] proposed an anchorage system in which the elastic modulus of the bonding medium gradually decreases from the free end of the anchorage to the loading end, as shown in Fig. 5a. The variable stiffness bonding medium was created by wrapping different thicknesses of epoxy resin layers around the outer surface of steel balls. This design effectively reduces the stress concentration in CFRP cables, ensuring a more even distribution of forces. The system was successfully implemented in the Stork Bridge, demonstrating its practical effectiveness in real-world applications. Wang et al. [44] proposed a new type of conical anchor featuring a novel continuous-fiber-reinforced load transfer component (LTC), as shown in Fig. 5b. This anchor achieves the effect of variable stiffness bonding medium by altering the winding angle of continuous fibers in the resin matrix. The results demonstrate that the variation in LTC modulus can effectively avoid shear and compressive stress concentration. As shown in Fig. 5c, Zhou et al. [45,46] created a variable stiffness bonding medium by adding short fibers to epoxy resin, finding that the compressive strength and elastic modulus increased with fiber content. Tensile tests on a cable supported by 37 BFRP rods showed an ultimate load of 766 kN, achieving an anchoring efficiency of 101 %, compared to the nominal rupture load of 761 kN. In conclusion, this type anchorage effectively mitigates stress concentration at the loading end, achieves progressive load transfer, and significantly enhances anchoring efficiency and durability.

2.2.4. Anchorages for CFRP cables

FRP cables are more challenging to anchor than FRP rods due to their greater flexibility and smoother surface, which makes it harder to establish a strong and reliable bond with the anchorage. However, FRP cables can be bonded into a cohesive force-carrying system using bonding media, making bonded-type anchorages the most common anchorage form for FRP cables [47]. Bonded-type anchorages have been widely used in FRP cable bridge structures both domestically and internationally [47,48].

The Neigles CFRP Footbridge uses epoxy resin bonded-type anchorages to anchor CFRP cables, with each cable consisting of 16 parallel CFCC 7-wire strands, and a total load-bearing capacity of 2272 kN. In this anchorage, the strand is bonded to the terminal using a special resin with an anchorage length of approximately 13.5 times the strand diameter, as shown in Fig. 6a [41,48,49]. The Herning CFRP Bridge uses epoxy resin to anchor CFRP strand cables. These CFRP cables, manufactured by Tokyo Rope Co., are CFCC 37-wire strands with a diameter of 40 mm, each capable of bearing a load of 1070 kN. The anchorage system is mainly composed of a straight tube shaped steel socket and a special resin filling, as shown in Fig. 6b. The anchorage length is approximately 13.5 times that of the cable diameter [48,50,51]. The pedestrian bridge at Jiangsu University uses a straight plus inner cone barrel anchorage design, where epoxy resin or expansive cement is injected into the barrel to anchor three types of CFRP cables with different numbers of strands, as shown in Fig. 6c. The load-bearing capacities of the tendons are 720 kN (6×8 mm), 1320 kN (11×8 mm), and 1920 kN (16×8 mm),

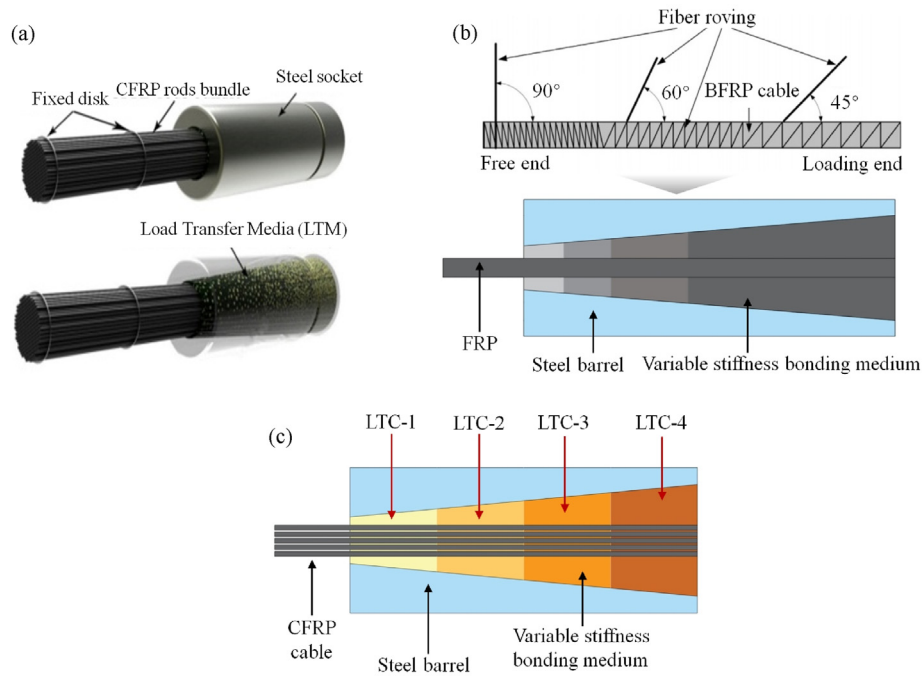


Fig. 5. Variable stiffness bonded-type anchorages with the media consisting of (a) Steel ball plus epoxy resin (b) Continuous-fiber-reinforced and (c) Epoxy resin plus short fibers [42,44,45].

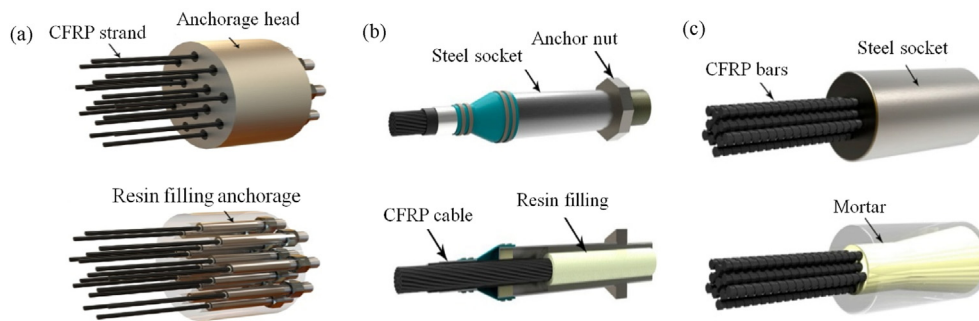


Fig. 6. FRP cable anchorages used in (a) Neigles CFRP Footbridge, (b) Herning CFRP Bridge and (c) Jiangsu University pedestrian bridge [41,48,52].

respectively [52–54].

In summary, bonded-type anchorage offers several advantages, including the prevention of localized shear damage, superior corrosion resistance, excellent fatigue performance, and reduced weight. These characteristics make it particularly suitable for applications in corrosive environments or where weight reduction is critical, such as in marine structures or long span bridge structure. However, this anchorage system also presents certain limitations: significant slip reduced load-bearing capacity due to the incorporation of bonding medium, and poor long-term creep resistance [55,56].

2.3. Composite-type anchorages

The composite-type anchorage is formed by combining the wedges and bonding medium either along or perpendicular to the fiber direction. This type of anchorage typically consists of FRP, anchor barrel, wedges and bonding medium. It relies on chemical bonding forces, mechanical interlocking forces and frictional forces to balance external loads [57–59].

Meng et al. [60] conducted experiments that demonstrated the shared tensile force on CFRP tendons by the steel tube and the bonding medium, which provides buffering and protection. Additionally, the combination of the three components disperses the local shear stresses on CFRP

tendons at the anchorage area, preventing shear damage caused by concentrated shear stress overload. Zhuge et al. [61] proposed a composite anchorage with bonding medium at the loading end of the anchorage and steel wedges at the free end, as shown in Fig. 7a. The free end experiences higher radial compressive stress, improving the anchoring effect and reducing stress concentration at the loading end to prevent shear damage. However, full clamping occurs only with more slippage at the anchorage, increasing the risk of bond failure. Additionally, Kim et al. [62] proposed a similar anchorage system, which successfully anchored CFRP bars with a diameter of 9.8 mm.

Mei et al. [63] designed a new parallel composite-type anchorage, as shown in Fig. 7b. The anchor barrel of this anchorage has a slight inclination difference of 0.1° from the wedges, creating a small gap near the loading end to reduce stress accumulation. Sun et al. [64] found that the tilt angle difference effectively avoids stress concentration, but increasing the tilt angle difference may lead to slip generation. Cai et al. [65] found that pre-tightening can improve the internal fitting of the anchorage system to some extent.

In conclusion, combined-type anchorages offer the advantages of reducing stress concentration and minimizing slippage. However, they are characterized by complex design and challenging manufacturing processes. Future efforts should focus on further optimizing the anchor design to address these limitations.

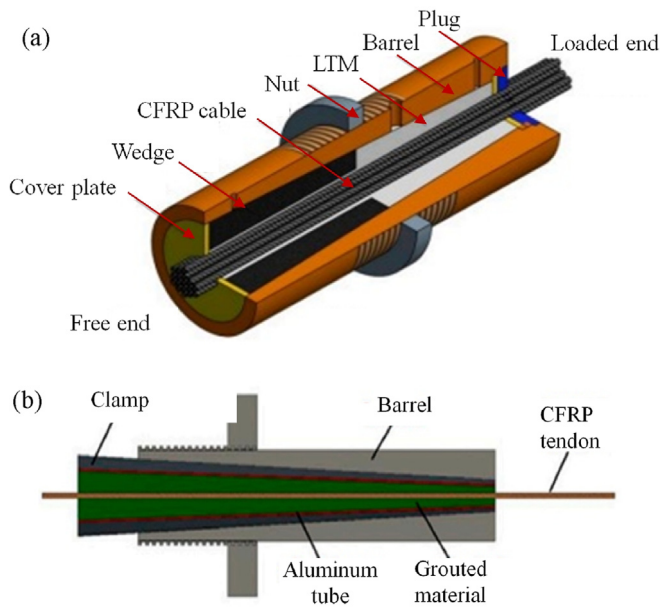


Fig. 7. Composite-type anchorages of (a) traditional tandem type and (c) anchorage with an angle difference [61,63].

2.4. Self-anchored systems

Self-anchored systems are a type of anchorage formed during the FRP fabrication process at its ends, relying on the tensile properties of the end material to balance external loads.

Zwingmann et al. [66] proposed an alternative design for pin-loaded straps (the sling anchorage) that increases the load bearing capacity up to 100 %, as shown in Fig. 8a. The experimental results show that the sling anchorage can achieve the predicted load bearing capacity. Furthermore, a manufacturing method that allows forming the loop anchorage at the ends of parallel CFRP tapes (or pultruded profiles) is also introduced. Ai et al. [67] proposed a novel design for self-anchored CFRP cables, which featured various anchorage configurations, as shown in Fig. 8b. The main variables in the study included the radius, thickness, and slope of the CFRP loop at the anchorage points. The results demonstrated that the mechanical behavior of the self-anchored cables is primarily influenced by the thickness and radius of the CFRP loop. By increasing these parameters, the material strength of the cable can be fully utilized. With optimized anchorage configurations, the anchor efficiency of the proposed self-anchored CFRP cable system can reach 95 % or higher.

In conclusion, self-anchored systems eliminate the need for complex installation procedures, offering advantages such as lightweight design, compact size, and ease of installation. However, they are prone to stress

concentration in the anchorage zone, which can lead to localized fracture and failure.

2.5. Summary of FRP anchorages

The primary types of FRP anchorage systems exhibit distinct characteristics in engineering applications. Mechanical-type anchorages transfer loads through clamping force, offering the advantages of easy installation and detachability, but are prone to localized stress concentration leading to surface damage on FRP, and may experience clamp relaxation issues during long-term service. Bonded-type anchorages rely on adhesives to achieve stress transfer, with benefits including uniform stress distribution and high anchorage efficiency, but are sensitive to construction precision and curing environment, and are susceptible to adhesive creep or interface aging under hot and humid conditions. Composite-type anchorages integrate mechanical clamping with adhesive mechanisms, offering high anchorage efficiency and long-term stability; however, increased structural complexity leads to higher costs, and multi-material interfaces may experience coordinated deformation issues under temperature load. Self-anchored systems utilize the inherent deformation of FRP materials for anchorage, featuring high degrees of structural lightweighting and no risk of metal corrosion, but require strict design matching with the creep characteristics of FRP, with anchorage performance prone to degradation under dynamic loading environments. Furthermore, current research trends are moving towards intelligent FRP anchorage systems, incorporating active control elements such as sensors and shape memory alloys to address the issues associated with traditional anchors.

3. Mechanical properties of anchorages

Understanding the stress distribution and deformation characteristics within each component of FRP anchorage system is essential for optimizing anchor design. While tensile testing alone cannot directly reveal the mechanical properties of the anchorage, researchers employ theoretical analysis and finite element simulations to assess the mechanical performance of the anchorage zone. These approaches provide comprehensive insights into the system's behavior under load, facilitating the enhancement of anchor design and performance without relying solely on extensive physical testing.

3.1. Theoretical analysis

The theoretical analysis of anchorage systems based on elastic-plastic theory is a crucial method for understanding the stress and deformation patterns in the anchoring zone, playing a significant role in optimizing anchor design. Zheng et al. [65,68] developed a numerical method to predict the nonlinear pull-out response of FRP rods in steel tubes filled

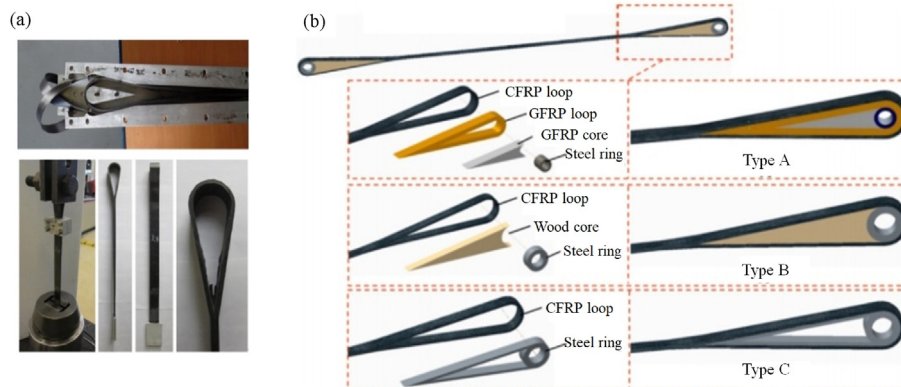


Fig. 8. Self-anchored systems of (a) pin-loaded straps type and (b) cone and ring type [66,67].

with cement grout, considering bond stress-slip behavior and unloading effects. The method uses 2 s-order differential equations based on the displacements of FRP rod and steel tube, and can also apply to FRP rods in rock when the steel tube is thick. After validation with experimental data, the method analyzes how the FRP rod's radius, length, bond stress, and frictional strength affect the maximum pull-out load. Ai et al. [67,69,70] conducted a theoretical analysis of the self-anchored anchorage based on an orthotropic thick-walled cylindrical model. The study revealed that the bending effect causes an uneven stress distribution across CFRP cable cross-section at the steel ring vertex. It was found that as the radius of the steel ring decreases, the stress unevenness becomes more pronounced.

In conclusion, the theoretical analysis research of FRP anchorage systems comprehensively examines the interfacial bonding-sliding mechanism, optimization of anchorage length and nonlinear characteristics of stress distribution. It proposes multi-scale design criteria based on fracture mechanics and elastoplastic models, providing quantitative design basis for enhancing the durability and load-bearing capacity of FRP-reinforced structures.

3.2. Simulation analysis

Finite element simulation analysis is the most commonly used method for indirectly obtaining the stress distribution and deformation behavior in FRP anchorage zone. By optimizing parameters such as anchorage length, cone angle, and material stiffness, predicting interface slip behavior and simulating complex stress fields, this method significantly enhances the reliability of anchor design and reduces testing costs [71–73].

Heydarinouri et al. [74] studied a curved clip-type anchorage system under a preload of 50 kN and a tensile load of 120 kN. They found that when the wedge enters the barrel, it tends to separate from the casing at

the loading end, which in turn reduces the contact pressure on CFRP rods, as illustrated in Fig. 9a. Duo et al. [28] conducted a finite element simulation analysis of prestressed mechanical-type anchorages, with the results shown in Fig. 9b. The peak transverse compressive stress slightly increased as the waveform number increased. Factors such as a higher gradient preload, shorter anchor length, thicker ear plate, and clamping plate helped distribute the axial stress more uniformly. Conversely, a longer anchor length, thicker ear plate and clamping plate, along with a thinner clip, were found to reduce the peak transverse compressive stress. Peng et al. [75,76] conducted a simulation analysis of variable stiffness bonded-type anchorages. As shown in Fig. 9c, the study indicated that the maximum Tsai–Wu failure index (FI_{max}) was linearly related to both the inner cone angle and the thickness of LTC at the loading end. An extremely low stiffness of LTC at the loading end can cause a rebound in the maximum FI_{max} due to the large interlaminar shear stress in FRP cable. Ai et al. [67] established a finite element model of self-anchored anchorages. As shown in Fig. 9d, the results indicate that the interfacial behavior between CFRP loop and steel wedge has an influence on the mechanical behavior of the anchorages. A full contact interface can lead to a favorable stress distribution of the anchorages. Increasing the radius and thickness of CFRP loop can effectively improve the load capacity of CFRP loop in a certain range. The critical values for the radius and thickness of CFRP loop for ensuring cable body failure are theoretically derived.

In conclusion, simulation analysis of FRP anchorage systems, through high-precision 3D modeling and multi-parameter coupled calculations, can efficiently analyze the nonlinear behavior of interfacial bond-slip, stress redistribution under dynamic loads and the evolution of local damage. This provides quantitative decision support for predicting structural failure thresholds, optimizing anchorage geometric parameters and enhancing the reliability of reinforcement.

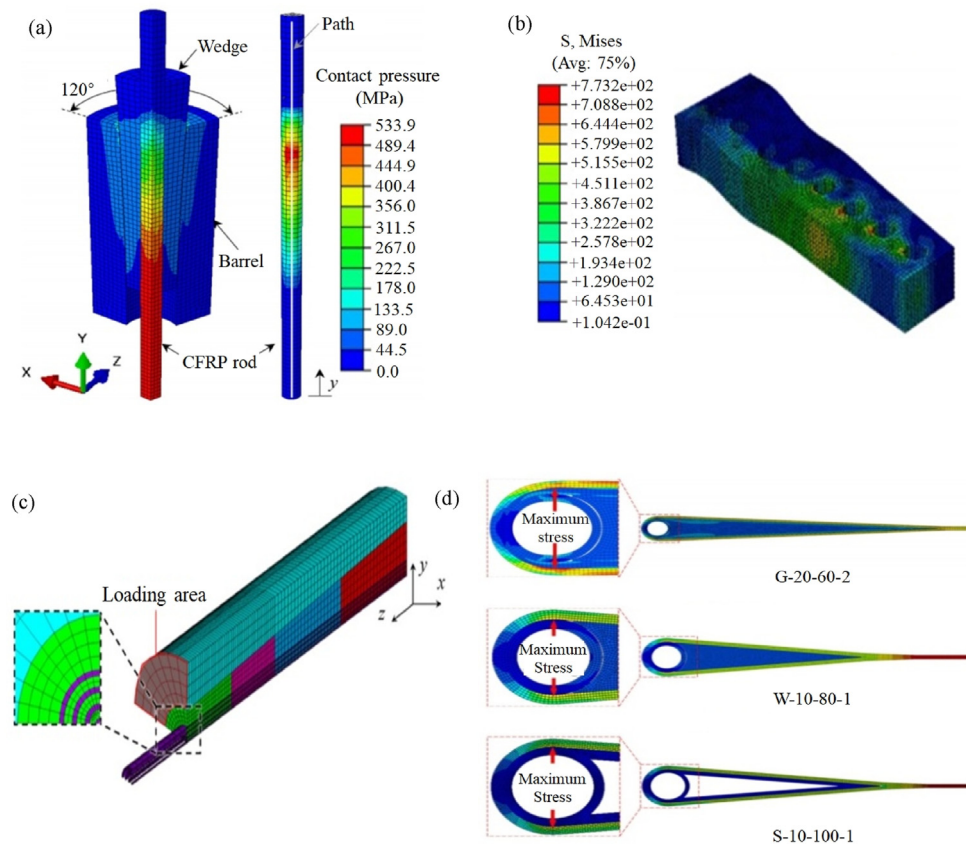


Fig. 9. Finite element simulation analysis of anchorages [28,67,74–76]: (a) curved clip-type anchorage system; (b) prestressed mechanical-type anchorages; (c) variable stiffness bonded-type anchorages; (d) self-anchored anchorages.

Theoretical analysis of FRP anchors enables a thorough exploration of stress distribution, deformation mechanisms, and failure modes through numerical simulations and theoretical models, providing a solid foundation for design optimization while reducing reliance on extensive physical testing, thereby saving costs and time. However, its limitations lie in the fact that the accuracy of theoretical models heavily depends on the precision of material parameters and boundary conditions, and it often struggles to fully replicate real-world complex conditions, potentially leading to discrepancies between analytical results and actual performance. Consequently, theoretical analysis is typically complemented by experimental validation to refine and verify the models, ensuring their reliability and practical applicability.

3.3. Damage mechanism of anchorages

The study of the damage mechanisms of FRP anchorages is crucial, as it directly influences the anchorage system's load-bearing efficiency, long-term stability and structural safety. This research serves as the core basis for preventing failure modes such as interface sliding, fiber cutting, and stress concentration. The following section will describe the damage mechanisms of the anchorage types mentioned above.

Mechanical-type anchorages can experience various forms of damage. Fracture damage occurs when the wedges' angle is too steep, causing excessive compression on FRP and leading to fracture. This can be avoided by adjusting the wedge angle and protecting FRP with aluminum or copper tubing [77]. Slip damage includes soft slip, which happens with low tensile stress and poor design, and dynamic slip, which occurs with high tensile stress. To prevent this, the friction coefficient, contact area, and normal pressure can be increased [78]. Fiber cutting damage happens when the wedge edges cut into FRP, causing fiber shearing and localized fracture. This can be prevented by using wedges with rounded edges or angle variations [14,79].

Bonded-type anchorages can experience two failure forms: shear failure and slip failure. Shear failure typically happens near the loading end, caused by excessive radial compression [33,80]. Slip damage involves movement between FRP and bonding medium, or between bonding medium and anchor barrel. This is usually caused by weak bonding strength or insufficient stiffness of the bonding medium. For instance, a short anchorage length can lead to insufficient bonding strength, causing cracks and reducing friction. Weak bonding media stiffness can cause deformation beyond the elastic range, resulting in anchorage failure. Applying pre-tension to FRP or using improved bonding media can enhance bonding and reduce slipping [81,82].

Composite-type anchorages reduce radial compressive stress on FRP, with slip damage being the main failure mode. In series composite-type anchorages, when tensile force is low, bonding and friction between the bonding medium and FRP bear the load. As force increases, bonding strength decreases, and the wedges at the free end transfer peak stress to the free end. In parallel composite-type anchorages, the wedges provide continuous compression, increasing radial forces as the tensile force grows [83,84]. Slip damage is affected by FRP surface, bonding medium properties and inner barrel shape. Threaded FRP reduces slip, while bonding media with high compressive strength and good deformation resistance also help. Inner cone barrels are more effective than straight barrels at reducing slip [85].

The damage mechanism of self-anchored systems is mainly characterized by fiber failure due to stress concentration, which is often caused by design defects. For example, excessive bending angles of FRP in the anchorage zone can lead to localized fracture damage in that area. This can be prevented by reinforcing FRP in the anchorage zone and reducing the bending angle of FRP.

4. Fatigue performance of anchorage systems

The fatigue performance of FRP anchorage systems is influenced by various factors, such as the type of anchorage and the specific parameters of the anchorage design. Additionally, test parameters play a critical role in determining the anchorages' endurance. These factors interact to affect the overall fatigue life, which is crucial for ensuring the long-term performance and reliability of FRP anchorage systems [86–88].

4.1. Fatigue performance of anchorages

Based on the most common types of anchorages, this paper summarizes research on the fatigue performance of mechanical, bonded and composite type anchorages, while noting that studies on the fatigue performance of self-anchored anchorages are relatively scarce.

Heydarinouri et al. [89] studied the fatigue performance of mechanical-type anchorages with camber contact surface between the wedges and the steel barrel. The fatigue tests revealed that the wedges seating distance significantly affected the occurrence of slippage between the components of the anchorage. In addition, at the loading frequency of 5–23Hz, there was no significant slip and temperature rise of the anchorage. Similarly, Elrefai et al. [86] found that the fatigue life of the tendon–anchor assembly was significantly affected by the stress range applied rather than the variation of the minimum stress; the fatigue life of the assembly decreased as the applied stress range increased.

Puigvert et al. [90,91] studied the fatigue properties of bonded-type anchorages with different bonding layer thicknesses. The results show that for the same normalised fatigue load, the sample with thinner bonding medium has higher fatigue cycles. Furthermore, Xie et al. [92] studied the fatigue performance of bonded-type anchorages under different stress amplitudes and loading frequencies. It was shown that when the stress amplitude is less than 10 % of the ultimate tensile stress and the maximum stress is less than 50 % of the ultimate bearing capacity, the tension-tension fatigue cycle can enhance the synergistic effect between the anchored components. In addition, a higher loading frequency leads to a sharp rise in temperature in the early and middle stages of loading, and a relatively stable temperature in the later stages.

Mei et al. [93] studied the fatigue performance of a composite-type anchorages. The results show that the fatigue cycle of the sample exceeds 2 million times at 45 % stress level. Fatigue failure in some samples is mainly due to interfacial dissociation between CFRP bar and the bonding medium. In addition, it is found that the fatigue performance of the sample with dispersed anchor is worse than that of the sample with parallel anchorage.

In conclusion, under cyclic fatigue loading, mechanical-type anchorages are prone to slippage and frequently exhibit stress concentration at their components. Bonded-type anchorages rely critically on the bond strength between FRP and grout, with their fatigue performance deteriorating as the bond degrades over time. Composite-type anchorages provide enhanced fatigue resistance through a synergistic combination of mechanical interlock and adhesive bonding. Nonetheless, they still encounter challenges associated with both slippage and bond integrity.

4.2. Fatigue damage mechanism of anchorages

The mechanism of fatigue damage in FRP anchorages is closely related to the anchorage type, and a detailed discussion will be provided in the following sections.

For mechanical-type anchorages, the fatigue damage mechanism primarily focuses on dynamic frictional wear at the contact interface and local stress concentration [77]. Under cyclic loading, repeated friction and compression at the contact area between the anchorage components and FRP lead to surface wear of the FRP and micro-crack propagation in the resin matrix [12]. Additionally, the difference in thermal expansion

coefficients between the metal anchorage and FRP exacerbates interfacial stress fluctuations. Moreover, geometric discontinuities in the anchorage tend to generate high-amplitude alternating stresses, causing progressive shear failure of fibers and plastic fatigue accumulation in the resin matrix [89]. This ultimately results in a reduction in the effective load-transfer area of the anchorage section and a decline in mechanical interlocking force. Studies suggest that optimizing anchor tooth parameters or adopting gradient stiffness designs can significantly mitigate stress concentration and extend fatigue life [77,94].

For bonded-type anchorages, the fatigue damage mechanism is mainly characterized by cyclic degradation of interfacial bonding performance [95]. Under cyclic loading, localized shear stress concentration at the interface between FRP and bonding medium initially induces micro-slip, leading to the initiation of micro-cracks along the ribs or fiber direction on the reinforcement surface. As the number of fatigue cycles increases, these micro-cracks gradually propagate and form interconnected damage zones, weakening the load-transfer efficiency of the interface [87,96]. Meanwhile, an increase in dynamic load frequency accelerates the slip rate. Furthermore, anchorage length and bonding agent stiffness significantly influence damage evolution: shorter anchorage lengths are prone to rapid failure dominated by interfacial debonding, while high-modulus bonding agents, although enhancing initial strength, may exacerbate fatigue damage accumulation due to brittle cracking. Research indicates that optimizing anchorage length or employing a flexible-rigid gradient bonding agent design can effectively delay interfacial damage propagation and extend fatigue life [92,97].

The fatigue damage mechanism of composite-type anchorages is primarily influenced by slippage between anchorage components and the degradation of the bond over time. This is caused by the interaction between the mechanical and bonding mechanisms, where the bond strength can weaken under cyclic loading [61,98].

For self-anchored anchorages, the fatigue damage mechanism mainly involves stress concentration and fiber failure. These issues arise from improper design, such as excessive bending angles or misalignment, which leads to localized fracture and reduced anchor performance under repeated loading [67].

5. Durability of FRP anchorage systems in complex environments

The fatigue performance of FRP anchorage systems is characterized by dynamic damage accumulation under cyclic loading, with failure mechanisms predominantly governed by stress concentration-induced crack initiation and propagation. In contrast, the durability primarily manifests as time-dependent material degradation and interfacial deterioration driven by environmental exposures (e.g., hygrothermal conditions, chemical corrosion), leading to progressive performance deterioration. In engineering practice, fatigue performance governs the lifespan design of vibration-sensitive structures (e.g., FRP cable anchorage zones in stay cables), while durability dictates the maintenance and design strategies for structures exposed to harsh environments (e.g., coastal FRP infrastructure). Due to the complex interactions between environmental factors and loading conditions, the long-term service performance of FRP anchorage systems faces challenges. This paper reviews the progress in research on the durability of FRP anchors, focusing on material properties, environmental factors, and interface degradation mechanisms [99–101].

5.1. Material properties and environmental factors

In complex service environments, the materials used in anchorage systems, such as FRP, bonding media, and steel sleeves, are susceptible to performance degradation. A detailed discussion on this issue will be presented in the following sections.

5.1.1. Durability of FRP

For FRP, the water absorption capacity of the resin matrix causes

water to diffuse within the material, while moisture absorption at the fiber-matrix interface weakens the bonding strength, ultimately leading to a loss of microstructural integrity. Moura et al. [102] studied the effect of hydrothermal aging on GFRP bars. Immersed in an alkaline solution (pH 12.6) for 1000 h at 23 °C and 60 °C, GFRP bars were evaluated for microstructural and chemical changes, moisture absorption, and mechanical property variations. The results showed increased voids and water absorption, reduced glass transition temperature of the resin, and changes in the glass fiber's chemical composition with aging. In a salt mist environment, Nan et al. [103] summarized the seawater aging mechanism of FRP composites, and the influence factors (matrix type, fiber type, immersion temperature, loading mode, and aging method) of seawater aging on the mechanical properties of FRP composites are further reviewed in detail, as shown in Fig. 10a. The results show that Cl^- ions penetrate the glass fiber (GFRP) and matrix interface through micro-cracks in the resin matrix, triggering an alkali-silica reaction (ASR) [104]. This leads to the formation of soluble silicate gel (such as Na_2SiO_3) on the fiber surface, causing fiber etching and a loss of strength. Additionally, the method for improving the seawater resistance of FRP composites is summarized include matrix modification, fiber surface treatment and composite surface coating, as shown in Fig. 10b.

Furthermore, in an alkaline environment, hydrolysis disrupts the epoxy-based interface chemical bonds of FRP, forming localized delamination-prone areas. The delamination strain threshold decreases to 40 %–50 % of the fiber fracture strain, and the extent of damage increases exponentially with higher OH^- concentrations. Feng et al. [105] investigated the tensile strength retention in GFRP bars of different diameters (13 mm and 25 mm) after immersion in an alkaline solution (pH = 12.6) at various temperatures (20 °C, 40 °C and 60 °C) for 1, 2, 3, and 6 months. The results reveal that the degradation of GFRP bars was slow at 20 °C, accelerated but not pronounced at 40 °C and considerable at 60 °C. Particularly, the 13 mm diameter GFRP bars exhibited a more significant reduction in tensile strength, with a decrease of 20.12 % after 6 months, while the 25 mm diameter bars only decreased by 13.23 %. M. Rifai et al. [106] conducted durability tests on BFRP bars in a high-temperature alkaline solution. The results showed that soaking in an alkaline solution at 60 °C for 6 months had a more detrimental effect on the microstructure and strength of BFRP than exposure to humid concrete. After 9 months of exposure at 60 °C, the specimens treated in the alkaline solution lost 29 % of their initial tensile strength.

5.1.2. Durability of bonding medium

Bonded-type anchorages typically use epoxy resin, cement-based materials, and reactive powder concrete (RPC) as bonding medium. However, under long-term and complex service conditions, the mechanical properties of the bonding materials degrade, directly affecting the reliability of the anchorage system and the service life of the structure.

Morgan et al. [107] found that in a hot and humid environment, the epoxy resin matrix is prone to hydrolysis, leading to molecular chain rupture and interfacial delamination. This effect is particularly pronounced under conditions of chloride ion or acid rain penetration, where bond strength may decrease by 30 %–50 %. This chemical degradation process accelerates with cyclic wet-dry conditions, as moisture diffuses through micro-cracks into the material, further exacerbating performance deterioration.

In addition, when RPC is exposed to high salt fog or acid rain environments for extended periods, it undergoes corrosion-induced expansion, leading to the formation of internal micro-cracks. After more than 1000 cycles, the expansion caused by freezing of pore water accelerates surface spalling. The bond interface between FRP and RPC may experience a 30 % reduction in bonding strength under hot and humid coupled conditions, and the slip displacement increases non-linearly with the load duration [108,109].

Furthermore, cement-based materials are prone to micro-cracking due to the expansion of pore water when frozen or the shrinkage stress

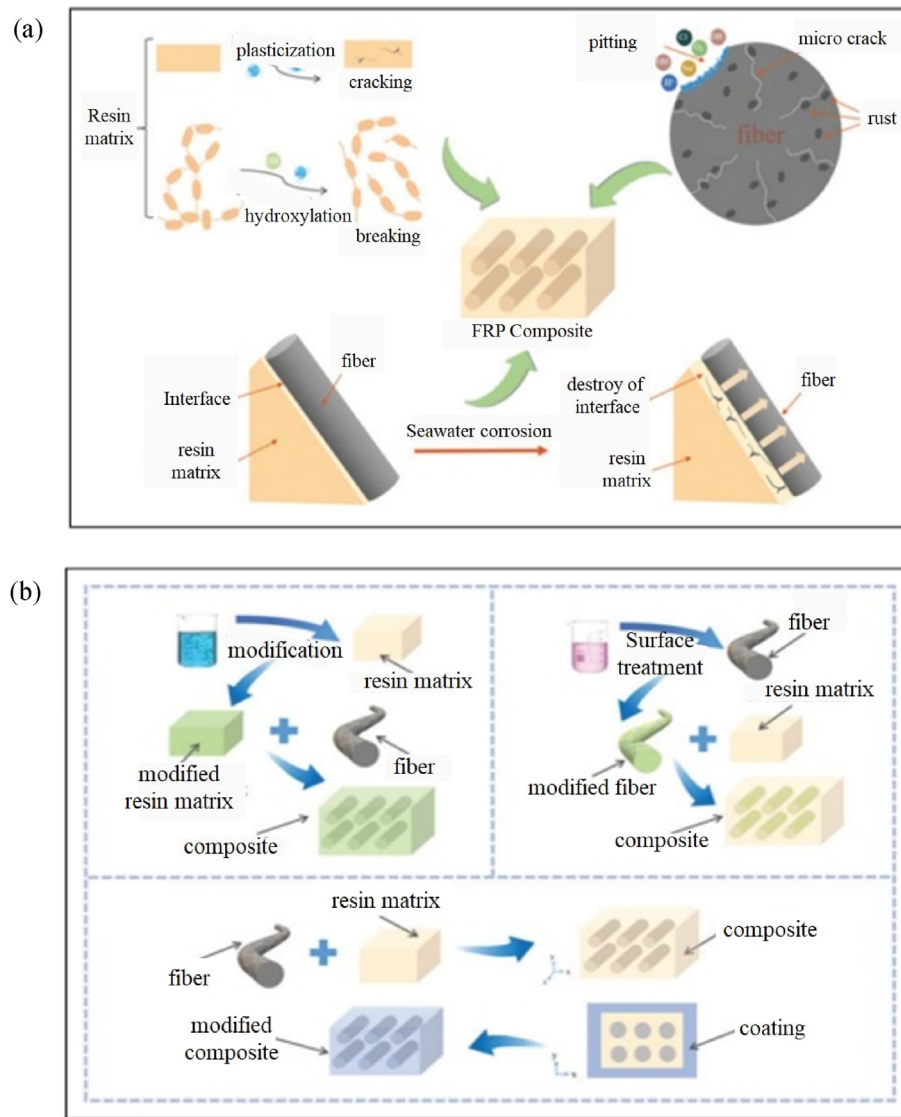


Fig. 10. Erosion damage mechanism and protection method of FRP [104]: (a) erosion damage mechanism of FRP, (b) matrix modification, fiber surface treatment and composite surface coating.

caused by dry-wet cycling. These micro-cracks can gradually develop into structural cracks, leading to a significant reduction in strength. Additionally, chloride ion penetration can induce corrosion-induced expansion, while sulfate corrosion reacts with cement hydration products to form expansive crystals, further accelerating the degradation of the material's performance [110–112].

5.1.3. Electrochemical corrosion of anchorages

Another key issue is the electrochemical corrosion in FRP mechanical-type anchorages in complex environments, mainly due to galvanic corrosion between the metal-fiber interface and localized corrosion from environmental media. When anchorages have metal parts, conductive CFRP and metal can form a potential difference in humid, salt or acidic environments, leading to rapid metal corrosion. Additionally, environmental media like Cl^- can penetrate micro-cracks in the resin matrix, causing pitting and stress corrosion cracking.

5.2. Interface degradation mechanisms

The durability of bonded or composite type's anchorages highly depends on the interface bond strength between FRP and the bonding

medium. The influence of environmental factors causes plasticization of the bonding medium near the interface, weakening its bonding with FRP and accelerating the accumulation of slip damage [113].

Silva et al. [114] conducted an experimental investigation on the reinforced concrete beams externally reinforced by CFRP or GFRP systems. The environmental corrosion, resulting from moisture, salt fog and temperature cycling, was investigated in their studies. It was observed that the salt water immersion of GFRP systems intensified the maximum bond stress which was associated to a corresponding shorter slippage. In contrast, no performance degradation occurred at the GFRP/concrete interface. Galvez et al. [115] evaluated the bond performance between CFRP and steel under durability conditions (60 °C, 99 % relative humidity) and thermal fatigue. The study revealed patterns of accelerated interface slip damage due to water and temperature, and uncovered the degradation mechanism of interface bond performance in a humid and hot environment.

5.3. Durability improvement solution

To improve the durability of FRP anchorages, multi-scale optimization measures are commonly applied in engineering. At the material

level, using hydrolysis-resistant resins (such as vinyl ester) or adding nano-fillers can enhance the aging resistance of both FRP and the bonding medium. In structural design, increasing the anchorage length helps distribute interface stresses, while gradient stiffness anchors effectively reduce local stress peaks. In terms of construction techniques, the combination of wet-sprayed polymer mortar and FRP rigid grids not only enhances the integrity of the anchorage system but also significantly extends its service life by avoiding structural adhesive aging issues. Additionally, smart monitoring technologies (such as embedded optical fiber sensors) can track interface damage evolution in real time, providing data support for durability assessment.

In summary, the durability research of FRP anchorages should further focus on the long-term degradation patterns under multi-field coupling effects. Integrated design combining materials, structure and monitoring should be employed to optimize performance throughout the entire service life. In the future, the development of standardized systems and the integration of smart maintenance technologies will be key areas for breakthroughs.

6. Conclusion

In this paper, based on static tensile properties, fatigue properties and durability properties of FRP anchorage system are systematically reviewed, and the conclusions are as follows:

- (1) The static bearing capacity of FRP anchorages is closely related to the anchorage form. The mechanical-type anchorages can significantly reduce the stress concentration effect by optimizing the geometrical parameters of the components and the friction coefficient of the interface. For bonded-type anchorage, the stress concentration can be avoided by changing the size of the steel barrel and the stiffness of the bonding medium. The composite anchorage can improve the anchoring efficiency and decrease the slip. But the existing anchors still have some disadvantages, such as large size and complicated fabrication.
- (2) The fatigue performance of FRP anchorage systems is primarily depended on the type of anchorage. Under cyclic fatigue loading, mechanical-type anchorages are prone to interface wear, local cutting, or sliding failure. Bonded-type anchorages may experience shear failure and sliding failure, while composite-type anchorages are predominantly characterized by sliding failure. Self-anchored systems are mainly affected by stress concentration. To mitigate fatigue damage, several strategies can be employed, such as adding protective casings, utilizing variable stiffness bonding media and optimizing the shape of the anchorage.
- (3) FRP and bonding media are susceptible to degradation when exposed to environmental erosion factors such as water, temperature, salt mist, and alkaline conditions. Additionally, electrochemical corrosion is likely to occur among the anchor materials. Furthermore, the interface bonding properties of each component within the anchorage system deteriorate under the combined effects of mechanical load and environmental exposure, significantly compromising the long-term durability of FRP anchorage systems.

CRedit authorship contribution statement

Yanzhao Niu: Writing – review & editing, Writing – original draft, Validation, Investigation, Data curation. **Guijun Xian:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Rui Guo:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation.

Declaration of interest statement

The authors declare that they have no conflict of interest.

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